



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

Enterprise Systems Design & Modelling (SECP3744)

Semester 1/20252026

Section 01

Assignment 1 (Technical Report)

Robotic Process Automation (RPA) in Enterprise Systems

Lecturer : Dr. Aryati Binti Bakri

Prepared by:

No.	Name	Matric Number
1	Lee Yin Shen	A23CS0236
2	Tan Zhi Ming	A23CS0189

Executive Summary

This report discusses Robotic Process Automation (RPA) as a solution for organizations to improve productivity by automating repetitive and manual tasks. Even in organizations with integrated Enterprise Resource Planning (ERP) system, they still need a lot of manual work, which can lead to errors and low data quality. RPA uses bots to emulate human actions to automate the business processes in digital systems. This report covers the main functions of RPA that include improving productivity, accuracy, and enhancing employee morale and its applications in Finance, Supply Chain, Healthcare, and Customer Services fields. Moreover, this report also discusses the challenges of RPA adoption, including employee resistance, complexity of scaling automation, and security concerns of data access. To solve these problems, this report recommends an organization to start with small projects, focus on change management, and establish a RPA team to manage the RPA strategy. This report concludes that RPA is a transformative technology, with its future trend "Intelligent Automation" through integration with AI that enables bots to handle more complex and decision based tasks.

1.0 Introduction

In today's competitive business environment, all organizations continuously find ways to improve their efficiency and reduce operational cost. A significant challenge for these companies is the large volume of repetitive and manual tasks that will consume too much time like copying data between spreadsheets and processing invoices. These processes are not only time-consuming but also highly lead to human errors that can negatively impact the data quality and decision-making in an organization. Even though the ERP system was designed to solve the problem of “information silos” by helping different departments share and access information from a single place, a lot of manual work is still required (Chen, J., 2022). Therefore, Robotic Process Automation (RPA) exists as a solution to address this problem.

This report provides an overview of RPA in enterprise systems, explaining how it functions, its practical applications, key advantages and challenges. The report also provides actionable recommendations for successful adoption and explores the future of RPA, highlighting its important integration with artificial intelligence.

2.0 Background / Literature Review

Initially in the 1920s, automation referred to manual and mechanical operations in industrial manufacturing. It wasn't until the 1990s that technological advances brought about workflow automation software for repetitive tasks like data entry and order processing. The combination of those technologies led to the emergence of RPA, a technology that simplifies programming while increasing accuracy and streamlines digital process automation (UiPath, 2024).

RPA is a type of technology that utilizes software bots to automate tasks typically performed by humans, such as digitizing data entry and document transfer. RPA can interact by using a GUI or an API, and will follow the preset rules to perform highly repetitive, logic-based tasks. An enterprise-level RPA system is normally built by components of the robot, Control Room, Developer Studio, and Integration Layer (UiPath, n.d.). RPA is categorized by three types, which are Attended RPA (Works alongside humans, triggered by user input or specific actions), Unattended RPA (Operates independently, without human involvement), and Hybrid RPA (Combines attended and unattended bots, allowing humans and bots to collaborate on processes). From a technical function perspective, the RPA mimics user interaction on software interfaces to execute pre-defined, rule-based workflows (UiPath, n.d.).

3.0 Analysis and Discussion

3.1 How RPA Technology Functions

RPA uses software bots that simulate human actions on the system. These bots work with applications by using the graphical user interface or Application programming interfaces. RPA platforms like UiPath, Automation Anywhere, and Blue Prism offer visual tools that let users create workflows using drag-and-drop features instead of coding. Each workflow outlines a series of actions, such as reading an invoice from an email, using Optical Character Recognition (OCR) to extract data, entering that data into an ERP system, and saving a confirmation report.

Bots fall into these categories:

- Attended bots assist human workers by taking care of repetitive parts of their tasks.
- Unattended bots operate on their own based on schedules or triggers.
- Hybrid bots mix both methods to provide flexibility and control.

The modern RPA system nowadays has integrated AI and machine learning features to enable them to make decision-making capabilities.

3.2 RPA Applications and Case Examples

3.2.1 Finance and Accounting (F&A)

In the financial sector, companies use RPA to process invoices, reconciling accounts, and generating reports. Besides, large enterprises that use ERP systems, such as SAP, deploy RPA bots to automatically perform tasks, such as extract data from supplier invoices without human involvement. Deloitte found that using RPA had cut the processing time for accounts payable tasks by 80% and reduced human error. Another popular application is compliance reporting. Bots collect and verify large datasets from different financial systems to prepare reports for audits, ensuring consistency and lowering the chances of manual mistakes (Li et al., 2025).

3.2.2 Supply Chain and Logistics (SCM)

RPA automates inventory updates, which enhances the visibility and data synchronization of the supply chain. Among global logistics companies, robots will monitor data from various systems and provide real-time updates to customers and suppliers. For example, a major logistics provider used RPA to automate order confirmations and shipping notifications, reducing response times from hours to minutes (Wewerka & Reichert, 2020).

3.2.3 Customer Service and IT Service Management (ITSM)

RPA assists customer service by automating processes such as ticket creation and data lookup in customer relationship management systems. In IT service management, robots perform tasks such as system monitoring and automatically triggering alerts as incident response. To free support teams to focus on more complex issues, IBM, for example, integrates RPA with its ITSM tools to automatically update users and handle low-level tickets (Wang et al., 2025). Additionally, customer service departments are using automated services such as those driven by chatbots, which combine technologies such as chatbots and RPA. In this service, AI manages customer interactions while RPA handles back-end tasks to fulfill customer requests.

3.3 Advantages and disadvantages

Advantages	Disadvantages
Increased Productivity: Bots can operate 24/7 without breaks, and also can complete repetitive tasks faster than humans.	High Initial Investment: The costs for software licenses, development, and the initial implementation can be quite high.
Improved Accuracy: By automating tasks like data entry, RPA can eliminate the human error that can lead to higher quality and reliable data.	Process Limitations: Bot is only suitable for tasks that are defined and standardized. It cannot handle works that are complex and need creativity.
Significant Cost Savings: Automating manual work can reduce operational costs, that can allow companies to achieve a higher Return On Investment (ROI).	Potential Employee Resistance: Some of the staff may worry that RPA will replace their jobs, which can lead to resistance and bring negative effects to the company culture.

Sources: Chavarria, A. (2022), *Canon Business Services ANZ*. (2024), *Automation Anywhere*. (2024)

3.4 Challenges and future trends

The adoption of RPA always faces challenges in an organization. One significant challenge is the **employee resistance**, where some of the employees will be afraid of job loss because of a lack of understanding of RPA (*Canon Business Services ANZ*, 2024). This will lower the employee morale and slow down the adoption of RPA. Another challenge is **process complexity and scalability**. As the company grows, they need to automate more advanced business processes. Therefore, it will be more complex, and the basic bots are unable to handle it (Chavarria, A., 2022). At the same time, more bots are needed. This combination of more complex tasks and growing number of bots need a strong plan and a powerful RPA system. One more challenge is the **security concern**. The bots need to always access the sensitive data of the company and customer to perform their tasks, which will cause data breach (*Automation Anywhere*, 2024).

In the future, the most **significant evolution** of Robotic Process Automation (RPA) is the integration with **Artificial Intelligence (AI)** that we call "**Intelligent Automation**" (*Dynamics Square USA*, 2025). This means a bot will not just follow simple and defined rules, but they can learn from data and make smart decisions. Another key trend is **cloud-based RPA**, which can make automation more

affordable and scalable for businesses by offering the service (RPAaaS) rather than using expensive on-premise infrastructure (Inamdar, N., 2025).

4.0 Recommendations

An organization needs strategic approaches to adopt RPA successfully.

First, organizations should **start with small projects**. Organizations can select some simple but high impact processes which are highly repetitive and rule-based. This will allow the RPA team to gain valuable experience and build confidence for the future programs. (Automation Anywhere., 2024)

Second, organizations should **focus on helping the employees with the RPA adoption**. They should communicate openly with employees, telling them that RPA is only a tool that will help them for repetitive work, and can free them up for more creative and strategic tasks. Moreover, organisations should provide training to enhance their employee skill to overcome the employee resistance (*Canon Business Services ANZ.*, 2024).

Lastly, organizations should **establish a RPA team** to manage the overall RPA operation. This team is important to propose RPA strategies and find the best plan to ensure all automation processes are scalable and aligned with the company business goals. This team is important to ensure the long term success and sustainability of the RPA process in an organization. (Automation Anywhere., 2024)

5.0 Conclusion

In conclusion, RPA is a powerful technology that can solve the continuous challenge of manual and repetitive works in an organization. It can increase productivity, enhance data accuracy, and reduce operational costs for an organization. A successful RPA adoption needs comprehensive strategy, including handling change management, addressing employee resistance, process complexity and scalability through clear communication and strong governance.

In the future, the integration of RPA and AI will define the next era of business process automation. This evolution will create a digital workforce that is able to handle more complex and decision-based tasks. Future research should focus on how this automation will affect employee skills and company culture over time, especially as people and bots work together more (Inc, U., 2016).

References

- Agentic business process management: The past 30 years and practitioners' future perspectives. (2015). arXiv. <https://arxiv.org/html/2504.03693v1>
- Automation Anywhere. (2024). *What is robotic process automation (RPA) | An enterprise guide*. <https://www.automationanywhere.com/rpa/robotic-process-automation>
- Canon Business Services ANZ. (2024). *Why you need RPA in your ERP systems*. <https://business.canon.com.au/insights/why-you-need-rpa-in-your-erp-systems>
- Chavarria, A. (2022). Robotic process automation (RPA): Definition and benefits. *Investopedia*. <https://www.investopedia.com/terms/r/robotic-process-automation-rpa.asp>
- Chen, J. (2022). Information silo? What it is, how it works. *Investopedia*. <https://www.investopedia.com/terms/i/information-silo.asp>
- Chugh, R., Macht, S., & Hossain, R. (2022). Robotic process automation: A review of organizational grey literature. *International Journal of Information Systems and Project Management*, 10(1), 5–26. <https://doi.org/10.12821/ijispm100101>
- Dynamics Square USA. (2025). Robotic process automation in manufacturing industry. *ERP Software Blog*. <https://erpsoftwareblog.com/2025/05/robotic-process-automation-in-manufacturing-industry/>
- Inamdar, N. (2025, April 7). *What is RPA as a Service (RPAaaS)? A guide for businesses*. Lucent Innovation. <https://www.lucentinnovation.com/blogs/it-insights/what-is-rpa-as-a-service-rpaaas>
- Li, J., Wu, L., & Zhou, Y. (2025). Research on the influence of RPA technology on enterprise risk management: A case of Western Securities. *Highlights in Business, Economics and Management*, 57, 65–71. <https://doi.org/10.54097/hc24rm87>
- Rejman, M. (2021, January 20). *40 real world use cases of robotic process automation (RPA) across industries*. Flobotics. <https://flobotics.io/blog/rpa-use-cases-across-industries/>
- UiPath, Inc. (2016, July 26). *The evolution of robotic process automation (RPA): Past, present, and future*. UiPath. <https://www.uipath.com/blog/rpa/the-evolution-of-rpa-past-present-and-future>
- Wang, Y., Huang, F., & Chiu, T. (2025). Robotic process automation and enterprise systems: Toward cost efficiency. *Journal of Emerging Technologies in Accounting*, 22(1), 119–136. <https://doi.org/10.2308/JETA-2022-024>
- Wewerka, J., & Reichert, M. (2020). *Robotic process automation – A systematic literature review and assessment framework*. arXiv. <https://arxiv.org/abs/2012.11951>

Appendix / Logbook

RPA Report Logbook					
Date	Activity	Tool use	Keyword/Prompt use	Reference	Person in charge
12 Oct	Discussed and selected the report topic (RPA) with teammate.	WhatsApp	N/A	N/A	LEE
13 Oct	Created report structure (table of contents). Distributed tasks for the reports to teammate.	Microsoft Word	N/A	N/A	LEE
15 Oct	Search for information about ERP and RPA. Write the introduction.	Gemini, Chrome	"Talk about ERP system and RPA." "What is RPA and how it applied in real-world application?" "Why an organization need RPA? Give simple explanation" "What is information silo?"	Chavarria, A. (2022). https://www.investopedia.com/terms/r/robotic-process-automation-rpa.asp Chen, J. (2022). https://www.investopedia.com/terms/i/information-silo.asp Canon Business Services ANZ. (2024). https://business.canon.com.au/insights/why-you-need-rpa-in-your-erp-systems	LEE
15 Oct	Conduct in-depth research on the background of RPA and read relevant literature.	ChatGPT, Google AI Mode	"Let's make a deep research regarding RPA, let's start with background section first. Please recommend relevant literature for me to have a better understanding for this RPA field."	Wewerka, J., & Reichert, M. (2020). Robotic process automation – A systematic literature review and assessment framework. <i>arXiv</i> . https://arxiv.org/abs/2012.11951 UiPath, Inc. (2016, July 26). <i>The evolution of robotic process automation (RPA): Past, present, and future</i> . UiPath. https://www.uipath.com/blog/rpa/the-evolution-of-rpa-past-present-and-future Wired. (n.d.). <i>How the pandemic accelerated RPA adoption (case examples)</i> . https://www.wired.com/story/pandemic-propelling-new-wave-automation	TAN
16 Oct	Explore how RPA operates and start writing about the background and how RPA functions.	Gemini, ChatGPT	"Please analyze the key points of the background information I provided and make up for any omissions." "Provide a detailed explanation of how RPA operates, delving into the technical aspects for a comprehensive understanding."	Chen, J. (2022). <i>Information silo? What it is, how it works</i> . Investopedia. https://www.investopedia.com/terms/i/information-silo.asp Automation Anywhere. (n.d.). <i>RPA overview + types of RPA</i> . https://www.automationanywhere.com/rpa/robotic-process-automation IBM Think. (n.d.). <i>RPA explanation and how it fits within business automation suites</i> . https://www.ibm.com/think/topics/rpa	TAN

18 Oct	Write advantages and disadvantages of RPA. Write challenges and future trends of RPA. Write recommendation for RPA adoptions.	Gemini, Chrome	"What are the advantages and disadvantages of RPA? Give real-world example in simple." "How about employee resistance? People will afraid to be replaced by robot right?" "What is Intelligent Automation" "Talk about Intelligent Automation and Cloud base RPA as the future trends of RPA in simple way."	Automation Anywhere. (2024). https://www.automationanywhere.com/rpa/robotic-process-automation Canon Business Services ANZ. (2024). https://business.canon.com.au/insights/why-you-need-rpa-in-your-erp-systems Chavarria, A. (2022). https://www.investopedia.com/terms/r/robotic-process-automation-rpa.asp Inamdar, N. (2025, April 7). https://www.lucentinnovation.com/blogs/it-insights/what-is-rpa-as-a-service-rpaas Wang, Y., Huang, F., & Chiu, T. (2025). https://doi.org/10.2308/JETA-2022-024	LEE
18 Oct	Understand the use cases of RPA in real-world scenarios, and complete section RPA Applications and Case Examples.	ChatGPT, Copilot	"In real life, what role does RPA play?" "How RPA Can Be Integrated with ERP in Real Life" "Please provide me with the roles that RPA plays in the financial, supply chain, and customer service industries."	Flobotics / ITRansition — <i>RPA use cases across industries</i>. https://flobotics.io/blog/rpa-use-cases-across-industries/ ERP Software Blog — <i>RPA in Dynamics 365 ERP</i>. https://erpsoftwareblog.com/2025/06/robotic-process-automation-in-dynamics-365-erp-on-autopilot/ Stampli — <i>RPA for accounts payable/invoice processing</i>. https://www.stampli.com/blog/all/robotic-process-automation-accounts-payable/	TAN
21 Oct	Complete executive summary, and conclusion part. Adjust the overall report format and pages. Add citation and appendix.	Gemini	"Summarize the whole report in a single and simple paragraph." "Write a short conclusion."	UiPath, Inc. (2016, July 26). https://www.uipath.com/blog/rpa/the-evolution-of-rpa-past-present-and-future	LEE
25 Oct	Do overall checking and submit the report.	N/A	N/A	N/A	LEE